Causal Inference Lab 03: Statistics

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Content

1. Causal Models
   - Average Treatment Effect (ATE)
   - Average Treatment Effect on the Treated (ATT)
   - Average Treatment Effect on the Untreated (ATU)
2. Estimators
   - Properties (Bias, Variance, Consistency)
3. Central Limit Theorem (CLT)

Motivation: Why Causal Models?

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- **All-Cause Model:**

$$Y = Y(S, U)$$

where S is treatment (policy, intervention, or exposure) and U is everything else (all other causes of Y).

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- **Ideally:** We would like treatment assignment (S) to be independent of all other factors (U).
- **More formally:**

$$S \perp U$$

  Treatment is independent of everything else that influences the outcome.
- This sets the stage for defining *causal parameters* like **ATE, ATT, ATU**.

Average Treatment Effect (ATE)

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$$\begin{aligned} ATE &= \mathbb{E}[Y(1,U) - Y(0,U)] \\ &= \underbrace{\mathbb{E}[Y(1,U)]}_{\substack{\text{Outcome if} \\ \textit{everyone} \text{ is} \\ \text{treated}}} - \underbrace{\mathbb{E}[Y(0,U)]}_{\substack{\text{Outcome if} \\ \textit{everyone} \text{ is} \\ \text{NOT treated}}} \end{aligned}$$

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- **The problem:** For any one person, we only observe *one* potential outcome.
    - If they are treated: we see $Y(1,U)$ but not $Y(0,U)$.
    - If they are untreated: we see $Y(0,U)$ but not $Y(1,U)$.

  This is the **fundamental problem of causal inference**.

Brief Detour

Causal Parameters

- **Population:** The entire group we want to study (e.g., all U.S. males).
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- **Sample:** A subset of the population used for analysis.
  - **Sample Statistic ($\hat{\theta}$):** A measurable value computed from the sample that estimates the population parameter (e.g., sample mean $\bar{x}$).

	Parameter (θ)	**Statistic ($\hat{\theta}$)**
Mean	μ	$\bar{x}$
Standard Deviation	σ	s

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- These parameters are **fixed characteristics** of the population, but we cannot observe them directly.
- The **Average Treatment Effect (ATE)** is also a population parameter:
  - It exists as a fixed number for the population.
  - But, like μ or σ^2, we must *estimate* it using data.
- **In practice:** The ATE exists as a fixed number, but we cannot observe it directly.
  - We use data + assumptions (e.g., randomization, $S \perp U$) to estimate it.
  - Example: In an RCT, the difference in average outcomes between treated and control is an unbiased estimator of the ATE.

Average Treatment Effect on the Treated (ATT)

Definition

$$
\begin{aligned}
ATT &= \mathbb{E}[Y(1,U) - Y(0,U) \mid S=1] \\
&= \underbrace{\mathbb{E}[Y(1,U) \mid S=1]}_{\substack{\text{Outcome of treated} \\ \text{(actually observed)}}} - \underbrace{\mathbb{E}[Y(0,U) \mid S=1]}_{\substack{\text{Counterfactual outcome} \\ \text{if treated had NOT been treated}}}
\end{aligned}
$$

- A **population parameter**, but restricted to the treated subgroup.
- **Interpretation:** Among those who actually received treatment, what is the average effect compared to if they had not been treated?
- **Challenge:** The second term involves a counterfactual, which we cannot observe directly.

Average Treatment Effect on the Treated (ATT)

Example: Job Training Programs

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  - $Y(1,U)$ = employment outcome if enrolled in training (observed for participants).
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  - $Y(1,U)$ = employment outcome if enrolled in training (observed for participants).
  - $Y(0,U)$ = employment outcome if *the same participants* had not enrolled (counterfactual).
- **Interpretation:** Among job seekers who received training, what was the average impact of the program on employment?

Average Treatment Effect on the Untreated (ATU)

Definition

$$\begin{aligned} ATU &= \mathbb{E}[Y(1,U) - Y(0,U) \mid S=0] \\ &= \underbrace{\mathbb{E}[Y(1,U) \mid S=0]}_{\substack{\text{Counterfactual outcome} \\ \text{if untreated had been treated}}} - \underbrace{\mathbb{E}[Y(0,U) \mid S=0]}_{\substack{\text{Outcome of untreated} \\ \text{(actually observed)}}} \end{aligned}$$

- A **population parameter**, but restricted to the untreated subgroup.
- **Interpretation:** Among those who did not receive treatment, what would the average effect have been if they had received it?
- **Challenge:** The first term involves a counterfactual, which we cannot observe directly.

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Example: Scholarship Programs (Education)

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  - $Y(1,U)$ = outcome (e.g., graduation rate) if the non-recipients had received scholarships (counterfactual).
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- **Interpretation:** Among students who did not receive scholarships, what would the average impact have been if they had?

Identification

(1) Parameters

- We've defined causal parameters like ATE, ATT, ATU
- These are quantities we'd like to know about the population

(2) The Problem

- Causal parameters involve counterfactuals (what would have happened under a different treatment status).
- This makes them unobservable directly.

(3) ***Identification***

Finding a way to express the causal parameter of interest (e.g., ATE) in terms of observable data and assumptions.

Estimators

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- **Example:** To estimate the population mean $\mu = \mathbb{E}[X]$, we use the **sample mean**:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

- Estimators themselves are **random variables** – their properties (bias, variance, consistency) help us judge whether they are "good".

Estimators

Why Properties?

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Why Properties?

- Not all estimators are created equal.
- Some estimators give results that are systematically off-target (**biased**).
- Some estimators jump around a lot from sample to sample (**high variance**).
- Some estimators improve as we collect more data, while others do not (**consistency**).
- **Motivation:** We want "good" estimators – ones that are accurate, stable, and reliable in large samples.

Properties of Estimators
(1) Bias

Bias

The bias of an estimator $\hat{\theta}_n$ for parameter θ is

$$\text{Bias}[\hat{\theta}_n] = \mathbb{E}[\hat{\theta}_n] - \theta$$

- If $\text{Bias}[\hat{\theta}_n] = 0$, the estimator is called **unbiased**.

Properties of Estimators
(1) Unbiased Estimator: Sample Mean

- **Example:** For $X_1, \ldots, X_n$ i.i.d. $\sim X$ with mean $\mu = \mathbb{E}[X]$, the sample mean
$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$
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$$\begin{aligned}
\text{Bias}[\bar{X}_n] &= \mathbb{E}[\bar{X}_n] - \mathbb{E}[X] \\
&= \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^{n} X_i\right] - \mathbb{E}[X] \\
&= \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}[X_i] - \mathbb{E}[X] \\
&= \frac{1}{n} \cdot n\mathbb{E}[X] - \mathbb{E}[X] \\
&= 0
\end{aligned}$$

Properties of Estimators

Biased vs. Unbiased Variance

1. Unbiased Variance:

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$$

Property:

$$\mathbb{E}[\hat{\sigma}^2] = \sigma^2$$

Interpretation: On average, this estimator equals the true variance.

2. Biased Variance:

$$\tilde{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$$

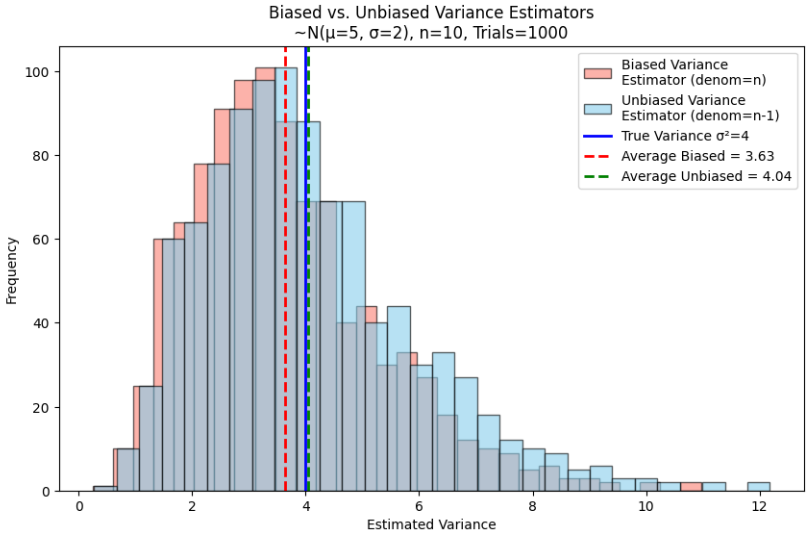
Property:

$$\mathbb{E}[\tilde{\sigma}^2] < \sigma^2$$

Interpretation: This estimator systematically underestimates the true variance.

Properties of Estimators

(1) Bias: Sample Mean – Visualization



Properties of Estimators

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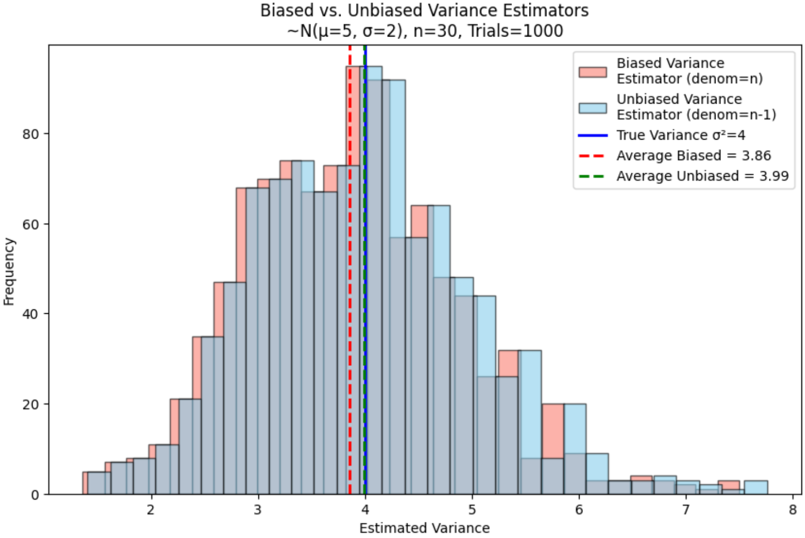

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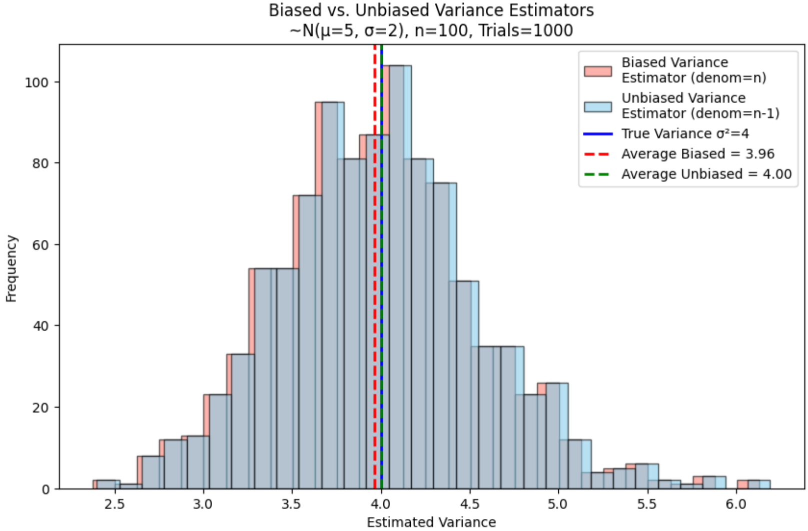

Properties of Estimators

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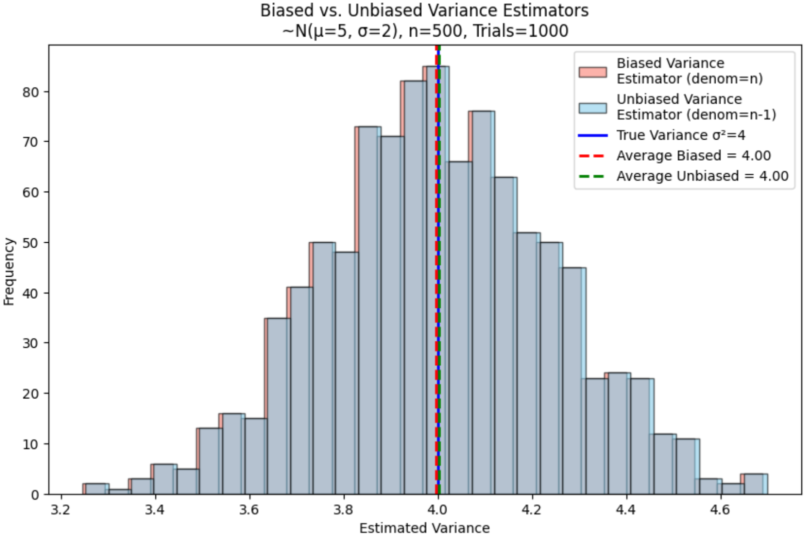


Properties of Estimators
(2) Variance

- **Variance of a random variable X:**

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right]$$

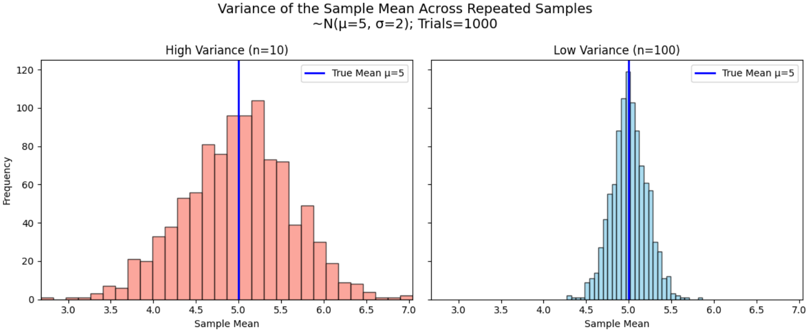
  - measures spread of the **data** itself.

- **Variance of an estimator $\hat{\theta}_n$:**

$$\mathrm{Var}(\hat{\theta}_n) = \mathbb{E}\left[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])^2\right]$$

  - measures spread of the **statistic** (e.g., sample mean, sample variance) across repeated samples.

Properties of Estimators
(2) Variance



- **Observe:** With small n, the distribution of the sample mean is wide (high variance). With large n, it becomes narrow (low variance).

Properties of Estimators

(3) Consistency

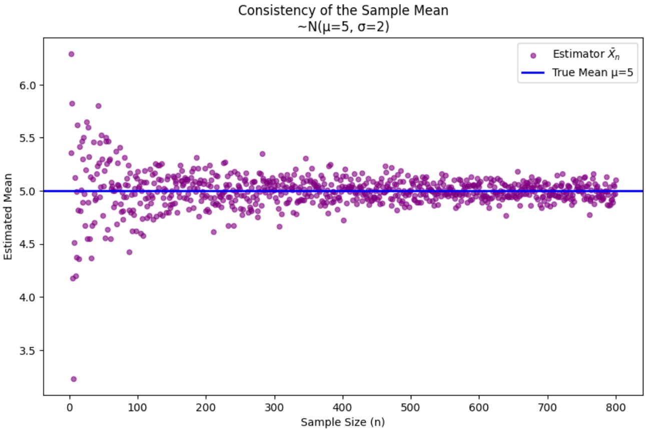
Consistency

An estimator $\hat{\theta}_n$ for parameter θ is **consistent** if, as the sample size grows,

$$P(|\hat{\theta}_n - \theta| > \varepsilon) \to 0 \text{ for any } \varepsilon > 0 \text{ as } n \to \infty.$$

- **In words**: as we collect more and more data, our estimator gets arbitrarily close to the true parameter.
- **Intuition:** The probability that our estimate is "off" by some amount ε shrinks to zero as n grows.
- **Example:** the sample mean $\bar{X}_n$ is a consistent estimator of $\mu = \mathbb{E}[X]$.

Properties of Estimators
(3) Consistency – Visualization



- **Observe:** As the sample size n increases, the estimator $\bar{X}_n$ clusters more tightly around the true mean $\mu = 5$.

Central Limit Theorem

Central Limit Theorem (CLT)

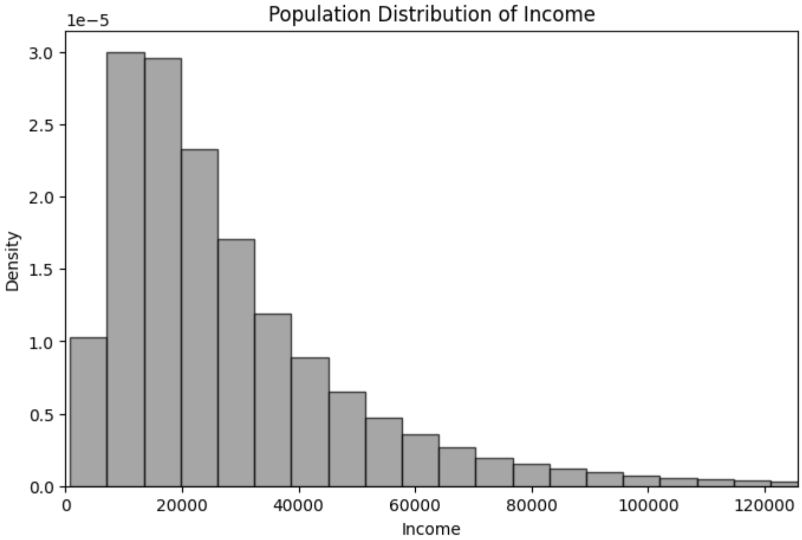
Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with mean μ and variance $\sigma^2 < \infty$. Then, as $n \to \infty$,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1).$$

- **Intuition:** Regardless of the original distribution of X, if we take large enough samples, the distribution of the sample mean looks approximately normal.

Central Limit Theorem

Example: Distribution of Income



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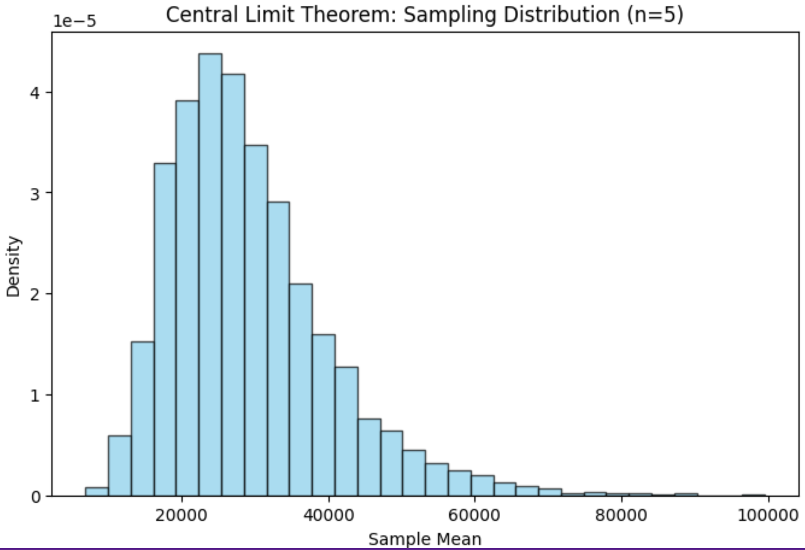

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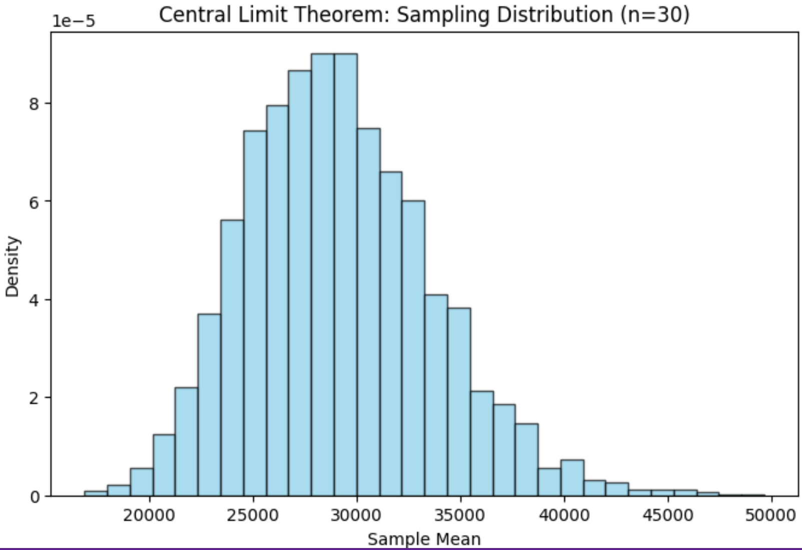

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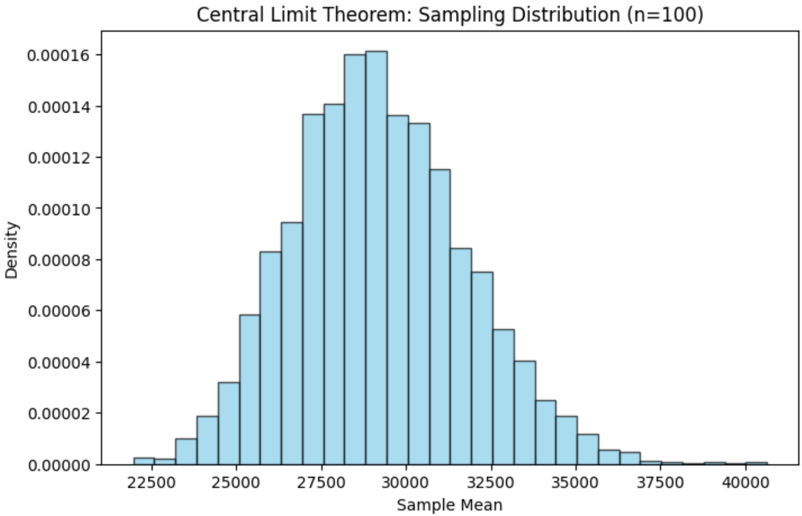

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